

How to Develop: Problem Statement, Topic, Objectives and Research Questions, Hypothesis

AN

A Seminar presentation

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27 February, 2023

PRESENTATION OUTLINE

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4. Information Systems Research

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2. AREAS OF RESEARCH IN ICT

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- Computer Science
- Information Systems
- **Software Engineering**
- Cyber Security,
ETC
- My main research area is Software Engineering/
Software Development
- SE is applied in all domains

3. SOFTWARE ENGINEERING RESEARCH

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- Model driven Engineering (MDE)
- Design using Unified Modelling Language (UML)
- System testing/Automated
- System Evaluation
- System maintenance and monitoring
- Automated software architecture and debugging
- Software defect prediction using machine learning

Definition of Concepts

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What is Software Engineering (SE) ?

- Software engineering is the process of applying engineering principles to analyze the needs of a specific user and then design, build and test software applications which will meet those needs.
- Software engineering goes beyond the fundamentals of programming to create more complex software systems that can serve as critical systems for businesses and organizations.
- Software engineers need to understand how to write software and how software and hardware interact.

What is Computer-aided software engineering (CASE)

- *Computer-aided software engineering (CASE)* uses programs to assist in the creation and maintenance of other computer programs.
- Programming has historically been performed by (human) programmers or teams; CASE adds software to the programming team.

Three (3) ways of finding a Problem Statement

- 1) Existing manual/traditional system is not working well
- (2) Existing automated/computerized system exist but giving problem.
- (3) A New system. No existing system, rather the project has not been done before in the community.

Sample Problem Statement

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SN	Project title	Sample problem statement
1	Development of Fake News and Advertisement Detection System	Individuals and groups have obligated themselves the task of spreading falsified information to other users with the aim of gaining some money, change the public's opinion on the integrity of reputable persons and organizations, potentially alter the results of an election or some other sinister reason.
2	Development of A Web-Based Medical Appointment System For Cove University	With the high number of appointments that are scheduled in clinics and hospitals every day, it is almost impossible for the clinic to achieve its goals. Definitely, there will be some dissatisfaction on both clinic and students. The clinic has to deal with urgent or emergency cases, absentees or no-shows among scheduled cases which may complicate the work of the receptionist who is booking these appointments manually through walk-ins.

3.1 FORMULATION OF PROBLEM (GAPS)

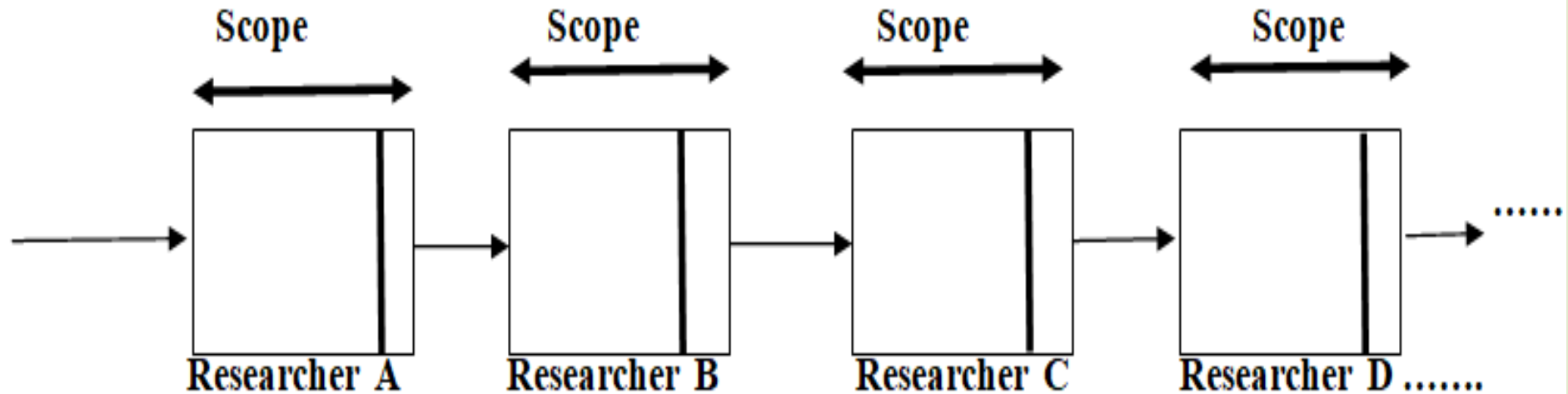
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- ➡ State the problem you want to solve
- ➡ Cite the publication/papers that contains the limitations of a study or a problem that you want to solve, and could not be solved in the current research
- ➡ Gaps are normally discovered through systematic review

1. INTRODUCTION

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RESEARCH CHAIN



Research is continuous (Research Advancements)

Some questions to address:

Which work are you extending?

What problem are you trying to solve?

What is the societal benefit of your research?

Etc

3.2 RESEARCH TOPIC

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- Must contain: a technique/methodology (**Using** or **a**), domain (in), system or platform(**of** or **for**),

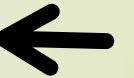
Examples

- Development of a model **for** result management system **in** E-Education **using** DevOps OR
- Development of **a** DevOps-based model **for** result management system **in** E-Education OR ←
- Development of **a** result management system **in** E-Education using DevOps

3.3 RESEARCH OBJECTIVE

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- The objective of this study is to develop a devops-based model for result management system in e-education



3.4 RESEARCH SUB OBJECTIVES

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The Sub **objectives** of this study are **To**:

- **To** carry out requirement elicitation of result management system
- **To** design a DevOps model for result management system
- **To** implement a result management system based on DevOps as proof of concept
- **To** evaluate the implemented system and benchmark with existing system

3.5 RESEARCH QUESTIONS AND HYPOTHESIS

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Research Questions

- How can a DevOps model be developed for result management system?
- How can a result management system based on DevOps be evaluated.

Hypothesis

- **H1:** The development of a DevOps model for result management system is realistic
- **H1:** A result management system based on DevOps has a high usability evaluation compared to other models.

3.6 RESEARCH METHODOLOGY

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- Tools/techniques such as software tools
- Research design/instruments such as questionnaire

3.7 CHAPTER OUTLINE OF DISSERTATION/THESIS

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A Sample Thesis outline:

- ➡ Ch1-Introduction
- ➡ Ch2-Literature review
- ➡ Ch3-Research methodology
- ➡ Ch4-Results and Discussion
- ➡ Ch5-Conclusion

3.8 SECTION OUTLINE OF PUBLICATION/PAPERS

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A Sample paper outline:

- Section1-Introduction
- Section2-Literature review
- Section3-Research methodology (System Design)
- Section4-Result and Discussion (System Implementation and Evaluation)
- Section5-Conclusion

4 INFORMATION SYSTEM RESEARCH

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Research Title: *IMPACT OF ICT ON STUDENT ACADEMIC ACHIEVEMENT*

Formulation of Research Questions and Hypothesis

- **Research Question 1:** *To what extent does teaching with the aid of ICTs impact on student academic achievement?*
- **Hypothesis H1:** *Teaching with the aid of ICTs has a positive impact on student academic achievement.*
- **Research Question 2:** *To what extent does course student engagement impact on their academic achievement?*
- **Hypothesis H2:** *Student engagement has a positive impact on their academic achievement.*
- **Research Question 3:** *To what extent does student use of ICTs impact on academic achievement?*
- **Hypothesis H3:** *Student use of ICTs has a positive impact on academic achievement.*

4.1 RESEARCH AIM

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The aim of this research is to explore the impact of ICT on student academic achievements

4.2 RESEARCH OBJECTIVES

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The objectives are as follows:

- To determine if teaching with the aid of ICTs impact on student academic achievement
- To determine the extent to which course student engagement impact on student academic achievement
- To determine the extent to which student use of ICTs impact on student academic achievement

4.3 MAPPING OF RESEARCH QUESTIONS, HYPOTHESIS AND RESEARCH OBJECTIVES

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SN	Research Questions (RQ)	Hypothesis (H)	Research Objectives (RO)
1	<i>To what extent does teaching with the aid of ICTs impact on student academic achievement</i>	<i>Teaching with the aid of ICTs has a positive impact on student academic achievement.</i>	<i>To determine if teaching with the aid of ICTs impact on student academic achievement</i>
2	<i>To what extent does course student engagement impact on their academic achievement?</i>	<i>Student engagement as a positive impact on their academic achievement</i>	<i>To determine the extent to which course student engagement impact on student academic achievement</i>
3	<i>To what extent does student use of ICTs impact on academic achievement?</i>	<i>Student use of ICTs has a positive impact on academic achievement</i>	<i>To determine the extent to which student use of ICTs impact on student academic achievement</i>

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS

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RQ1: *To what extent does teaching with the aid of ICTs impact on student academic achievement?*

Construct B: Teaching with the aid of ICTs returned a strong positive correlation and prediction (2nd in weight(β) out of 5 predictors) of *student academic achievement*.

H1: *Teaching with the aid of ICTs has a positive impact on student academic achievement.*

H1 was supported since a significant positive relationship was established with model 1 & 2. (Unstd β = 5.954 & 5.848, $p < 0.01$). The finding is in agreement with the results of Bai, et al. (2016); Rugutt and Chemosit (2005.); Ilomaki (2008); Abuet *al.*, (2011); Oye *et al.*, (2012) and Angeli&Tsaggari (2016).

4.4 CONFIRMATION OF RESEARCH QUESTIONS AND HYPOTHESIS (CON'T)

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RQ2: *To what extent does course student engagement impact on student academic achievement?*

Construct C: Student engagement, *had* a weak positive correlation with, and prediction (3rd in weight (β) out of 4 predictors) of Class Marks.

H2: *Student engagement has a positive impact on student academic achievement.*

H2 however, was supported since a significant positive relationship was established for model 1 & 2 (Unstd β = 1.086 & 1.140, $p < 0.05$). The finding is in line with Gunuc's (2014) conclusions.

5. PUBLICATION OF RESEARCH OUTPUT

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Publishers of good Journal Outlets

- Nature
- Springer Nature
- Elsevier
- Sage
- Emerald
- Wiley
- University-based Journals
- Etc

Conference outlets

- ACM
- IEEE
- Springer
- Etc

Index Outlets/Databases

- Scopus
- Web of Science/ISI
- University databases/
repositories
- Etc

5. CONCLUSION

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This presentations has no doubt achieved its objectives of highlighting the research methodology in Computer Science and related areas

THANK YOU